EduGenie – Google Gemini Powered Learning Assistant

Coding & Solution

Team ID: SWTID-2026-5557 **Date:** 30 September 2026

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Implementation Summary

EduGenie uses Python and FastAPI to expose academic assistance endpoints. Requests are validated, routed to feature-specific modules, converted into prompts, and passed to a shared Gemini Service Layer. The service layer discovers models available to the configured API key and can fall back to alternative models or an offline/demo explainer.

Feature Implementation

Module / Endpoint Group	Implementation Role
Q&A	Accepts an academic question and returns an AI-generated answer.
Explain	Accepts topic and level and returns a level-appropriate explanation.
Summarize	Accepts a passage and produces a concise summary.
Quiz	Accepts study material and requested question count; returns quiz structure and answer key.
Learning Path	Accepts a topic and produces an ordered, leveled learning roadmap.
Gemini Service	Handles API access, runtime model discovery and fallback.
Offline/Demo Mode	Keeps the application usable without a configured live API key or network.

Security and Configuration

Gemini credentials are loaded through environment variables using python-dotenv rather than being embedded in frontend code.